

Flowpilot Project Requirements Document

Version 1.0.0 | Date: July 15, 2026

This document outlines the architectural components and features required for the Flowpilot Server MVP. The goal of this project is to build an AI-powered project management platform that parses documents and automatically builds Work Breakdown Structures (WBS).

1. User Authentication System

The system requires a robust and secure authentication mechanism to handle tenant-based sessions.

- Design and implement a MongoDB/Beanie database schema for User and Tenant models.
- Expose FastAPI routes for **/auth/register** and **/auth/login**.
- Implement JWT token-based authentication using PyJWT.
- Secure passwords using the bcrypt library with salt generation.

2. Document Parsing & WBS Engine

The primary feature of Flowpilot is parsing unstructured text/requirements and generating a WBS.

- Build an upload endpoint to accept **.pdf**, **.docx**, and **.txt** files.
- Implement text extraction using **pypdf** for PDFs and **python-docx** for Word files.
- Integrate **mlx_lm** to load local LLMs on Apple Silicon (e.g. DeepSeek-R1-Distill-Qwen-14B).
- Stream WBS generation using Server-Sent Events (SSE) in JSON format.

3. Vector Database Storage

For semantic search and document retrieval, the system must index files.

- Initialize a LanceDB local store database in **.lancedb_store**.
- Implement text chunking and generate embeddings using standard HuggingFace models.
- Add vector search endpoints to search through parsed requirements.